UNIT I – Fundamentals of Generative AI and Large Language Models

1. Install and configure Python, Jupyter Notebook/Google Colab, and required AI libraries (NumPy, Pandas, Transformers, PyTorch).

Aim:

To configure the Python environment and install the libraries required for transformer-based AI experiments.

Algorithm:

Start the program.
Open Google Colab and create a new Python notebook.
Install the required libraries: NumPy, Pandas, PyTorch, and Transformers.
Import all the installed libraries into the Python environment.
Display the version of each library.
Check whether GPU is available; otherwise, use CPU.
Verify that all libraries are imported successfully.
Stop the program.

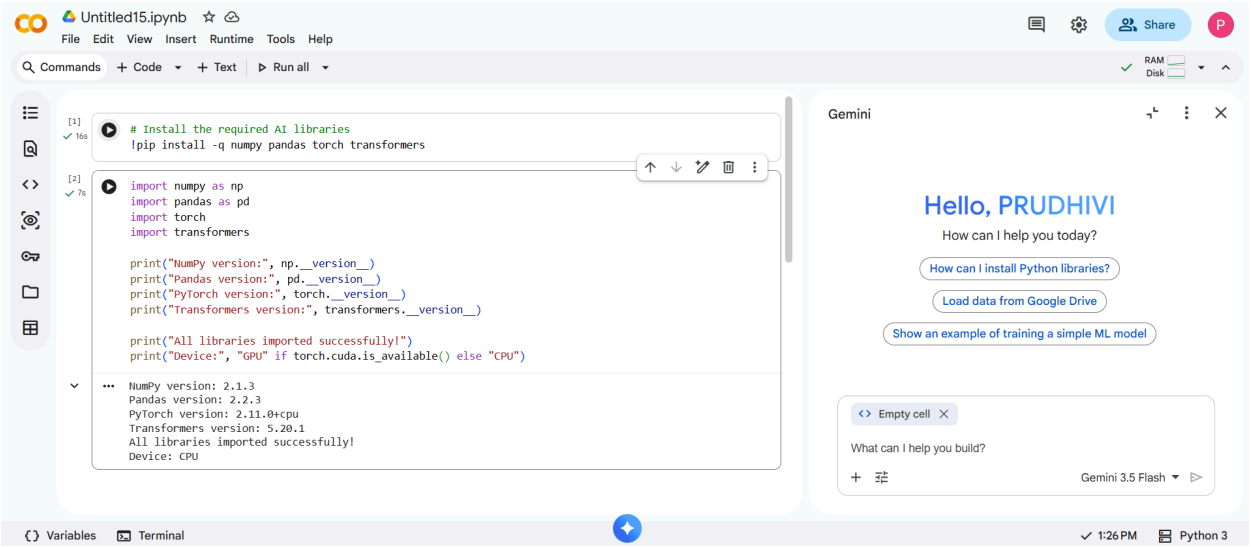
Code:

```
# Install the required AI libraries
!pip install -q numpy pandas torch transformers
import numpy as np
import pandas as pd
import torch
import transformers

print("NumPy version:", np.__version__)
print("Pandas version:", pd.__version__)
print("PyTorch version:", torch.__version__)
print("Transformers version:", transformers.__version__)

print("All libraries imported successfully!")
print("Device:", "GPU" if torch.cuda.is_available() else "CPU")
```

Screenshot:



Result:
Python environment configured successfully with the required AI libraries.